QUIZ #3 for Section 2

This is a 30-minute quiz

Student ID:	Student Full Name:

Write a Pandas program that processes the following JSON data representing customer orders.

```
orders_json = '''
[
    {"OrderID": 101, "Customer": "Ali", "City": "Istanbul", "Amount": 2500, "Status": "Completed"},
    {"OrderID": 102, "Customer": "Ayse", "City": "Ankara", "Amount": null, "Status": "Completed"},
    {"OrderID": 103, "Customer": "Mehmet", "City": "Izmir", "Amount": 1200, "Status": "Cancelled"},
    {"OrderID": 104, "Customer": "Zeynep", "City": "Istanbul", "Amount": 3100, "Status": "Completed"},
    {"OrderID": 105, "Customer": "Can", "City": "Bursa", "Amount": null, "Status": "Pending"},
    {"OrderID": 106, "Customer": "Elif", "City": "Ankara", "Amount": 1800, "Status": "Completed"}
]
'''
```

Your program must perform the following operations:

1. Import the required libraries.
2. Convert the JSON string into a Python object.
3. Convert the Python object into a Pandas DataFrame.
4. Print the first five rows of the DataFrame.
5. Print the number of missing values in each column.
6. Replace missing values in the Amount column with the median value of Amount.
7. Create a new column named HighValueOrder:
    - True if Amount >= 2000
    - False otherwise
8. Select and print only the completed orders where Status is "Completed".
9. Group the data by City and print the average Amount for each city.
10. Export the final DataFrame to a JSON file named: cleaned_orders.json